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EXAMINER
DIXON, ANNETTE FREDRICKA

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3785	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/112,559	Applicant(s) BARLOW et al.	
	Examiner ANNETTE F DIXON	Art Unit 3785	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 2/22/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 2/22/2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☑ Certified copies of the priority documents have been received in Application No. 17255615.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date ALL IDS.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Primary Examiner acknowledges Claims 1-20 are pending in this application as originally filed on February 22, 2023.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Drawings

3. The drawings are objected to for failing to comply with 37 CFR 1.84(q) because:

   a. The lead line for "3104" as shown in Figure 4P is missing. Appropriate correction is required.

4. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the

filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 103

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. Claims 1, 2, 5-7, 9, 10, 12-15, and 17-20 are rejected under 35 U.S.C. 103 as being unpatentable over Jablonski (2015/0128949) in view of Ovizinsky et al. (2014/0102456).

As to Claim 1, Jablonski discloses a patient interface (Figures 1 and 2), comprising: a plenum chamber (8, "a patient interface device 8 having a fluid coupling conduit 10 (e.g. an elbow conduit). ... Patient circuit 6 is structured to communicate the flow of breathing gas from pressure generating device 4 to patient interface device 8, and typically includes a gas delivery conduit or tube coupled to fluid coupling conduit 10." Para 0017) pressurizable (via 4 as connected to 8, wherein 4 – "pressure generating device 4" Para 0017) to a therapeutic pressure ("PAP", "Pressure generating device 4 is structured to generate a flow of breathing gas and may include, without limitation, ventilators, constant pressure support devices (such as a continuous positive airway pressure device, or CPAP device), variable pressure devices (e.g.,

BiPAP.RTM., Bi-Flex.RTM., or C-Flex.TM. devices manufactured and distributed by Philips Respironics of Murrysville, Pa.), and auto-titration pressure support devices." Para 0017), the plenum chamber (8) including a plenum chamber inlet (10, "a fluid coupling conduit 10 … and typically includes a gas delivery conduit or tube coupled to fluid coupling conduit 10." Para 0017) sized and structured to receive a flow of air (via 4, wherein 4 – "pressure generating device 4" Para 0017) at the therapeutic pressure ("PAP") for breathing by the patient, a seal forming structure (12, "an interface element 12, which in this exemplary embodiment is a nasal cushion" Para 0019; **also see:** "a interface element (12) structured to engage a portion of the face of a patient" Abstract) constructed and arranged to form a seal with a region of the patient's face ("interface element 12 is held in place over the nose of the patient so that gasses are permitted to be communicated to and from the airway of the patient through interface element 12" Para 0020) surrounding an entrance to the patient's airways (via "nasal cushion"), the seal forming structure (12) having a hole therein (defined by the region upon which the nose is received – "an interior space defined by interface element 12" Para 0019) such that the flow of air at the therapeutic pressure ("PAP") is delivered to at least the entrance of the patient's nares, the seal forming structure (12) constructed and arranged to maintain the therapeutic pressure ("PAP") in the plenum chamber (8) through the patient's respiratory cycle in use; a positioning and stabilizing structure (14, "Headgear assembly 14, described in greater detail below, is fluidly coupled to fluid coupling conduit 10 and provides a flow path that allows the flow of breathing gas generated by pressure generating device 4 to be communicated to an interior space defined by interface element 12 and then to the airway of a patient." Para 0019; **also see**: " a headgear assembly (14) coupled to the mask." Abstract; "headgear assembly 14 is

configured to secure patient interface device 8 to the patient's head in a manner wherein interface element 12 is held in place over the nose of the patient so that gasses are permitted to be communicated to and from the airway of the patient through interface element 12." Para 0020; "headgear assembly 14 is configured to secure patient interface device 8 to the patient's head when patient interface device 8 is donned by the patient." Para 0022) to provide a force ("secure" via frictional retainment – see Para 0020 and 0022) to hold the seal forming structure (12) in a therapeutically effective position on a head of a patient, the positioning and stabilizing structure (14) comprising: at least one gas delivery tube (20x, "Headgear assembly 14 includes a fluid delivery housing 20, described in greater detail herein, and first and second rear strap members 22A, 22B that extend from fluid delivery housing 20." Para 0021; **also see**: "Fluid delivery housing 20 includes a top surface member 28 and a bottom surface member 30, which are joined to one another along the outer perimeters thereof to form an airtight internal chamber 32." Para 0024) to deliver the flow of air to the entrance of the patient's airways via the seal forming structure (12), the at least one gas delivery tube (20x) being constructed and arranged to contact in use, at least a region of the patient's head superior to an otobasion superior of the patient's head (as shown in Figures 1 and 2); wherein the at least one gas delivery tube (20x) includes an inner textile layer (44, best seen Figures 3 and 4, "bottom surface member 30 comprises a two layer structure in the form of a urethane backed material and includes a fabric layer 44 attached to a urethane layer 46." Para 0030) and an outer textile layer (40, best seen Figures 3 and 4, "top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030), wherein the inner textile layer (44) and the outer textile layer (40) are joined (best seen Figure 4) along a first longitudinal edge at a first joint (one of

left or right of Figure 4) and along a second longitudinal edge at a second joint (other of left or right of Figure 4); wherein between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4) the outer textile layer (40) is spaced (via 42/46 at the joints and/or via 32/34 at the latitudinal axis of Figure 4, wherein 42/46 – "a urethane layer 42 … a urethane layer 46" and wherein 32/34 – "a plurality of spaced support members 34 are provided within internal chamber 32, with the top surface of each support member 34 being in engagement with the inner side of top surface member 28 and the bottom surface of each support member 34 being in engagement with the inner side of bottom surface member 30" Para 0024) from the inner textile layer (44); and wherein the outer textile layer (40) supports its own weight between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4) such that the outer textile layer (40) remains spaced from the inner textile layer (44) when not supported by pressurized air.

Regarding the concept of "supporting its own weight … remains spaced", **at the joints** (left or right of Figure 4) the intermediary structure (42/46) provides for a spaced arrangement between the outer textile layer (40) and the inner textile layer (44) – which effectively provides for the spaced arrangement wherein the weight of outer textile layer (40) is supported by its structural construction. Additionally, **at the points intermediate to the joints along the latitudinal axis of Figure 4 – explicitly central 32**, the construction of the intermediary structures (34) provides the central pathway (32) by which air is directed along the at least one gas delivery tube (20) – which effectively provides for the spaced arrangement whereby the central pathway (32) between the intermediary structures (34) are supported by the weight of the outer textile layer (40) to prevent collapsing of the at least one gas delivery tube (20) at the

central pathway (32). Still further, **at the points intermediate to the joints along the latitudinal axis of Figure 4 – explicitly lateral 32/32**, the construction of the intermediary structures (34) provides the lateral pathways (32) by which air is directed along the at least one gas delivery tube (20) – which effectively provides for the spaced arrangement whereby the lateral pathways (32) between the intermediary structure (34) and the joints (left or right of Figure 4) are supported by the weight of the outer textile layer (40) to prevent collapsing of the at least one gas delivery tube (20) at the lateral pathways (32). Consequently, as the claims do not preclude or prevent the inclusion of intermediate layers at the joint and/or intermediary structures along the latitudinal axis of the at least one gas delivery tube, it appears Jablonski meets the limitations of "supporting its own weight … remains spaced" as claimed.

Yet, Jablonski **does not expressly disclose** the explicit operational pressure values of the therapeutic pressure – to be "at least 6 cm of water above ambient pressure" **nor** the configuration whereby "wherein the outer textile layer is thermoformed to have a predetermined shape and the inner textile layer is not thermoformed"

Regarding operational pressures, Ovizinsky teaches an additional patient interface (Figures 6-15) suitable for imparting a therapeutic pressure, in the form of PAP to a patient, wherein "the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient interface via the air delivery conduit" (Para 0105). In light of the teachings of Ovizinsky the generation of therapeutic pressure of "at least 6 cm of water" was known as Ovizinsky considers the range of "2-30 cm H.sub.2O" which includes a range coextensive of with the claimed range of 6-30 cm of water was known in the prior art to provide PAP therapeutic pressure to a patient utilizing the patient interface.

Regarding the concept of thermoforming, Ovizinsky teaches an additional patient interface (Figures 6-15) suitable for having components which are thermoformed in the construction of the at least one gas delivery conduit (620, as best seen in Figures 8-15, "An air delivery conduit (620) includes first (621) and second (622) conduit portions that cooperate to form the conduit, each conduit portion including an inner layer of a film laminate (650) that forms an interior surface of the conduit and an outer layer of a textile (655) that forms an exterior surface of the conduit." Abstract). Explicitly, Ovizinsky teaches thermoforming is a known "method of processing flat material such as polyester or nylon into a finished three-dimensional shape." (Para 0141) whereby "an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit" (Para 0187) is constructed from "a thermoformable fabric" (Para 0187) such that "During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties." (Para 0188).

Although the construction of Ovizinsky requires thermoforming of both outer textile layer (655 of one of 621/622) and the inner textile layer (655 of the other of 622/621) to produce a circular shaped conduit, there is no structure that would preclude, prevent, or hinder the consideration of thermoforming as claimed whereby the outer textile layer (40) of Jablonski is thermoformed; whilst, the inner textile layer (44) of Jablonski is not thermoformed, to produce the predetermined shape of Jablonski's tube as shown in Figure 4 which includes the inner textile layer (44) in a planar configuration abutting the face of the patient and the outer textile layer (40) is in an arcuate shape as exposed to the ambient atmosphere. As the arcuate shape of the outer textile layer (40) of Jablonski is effectively a three dimensional shape, the decision to "processing flat material ... into a finished three-dimensional shape" (Para 0141) as

taught by Ovizinsky through thermoforming would be obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the configuration of the outer textile layer of the modified Jablonski having an arcuate shape through processing of thermoforming, whilst the inner textile layer of the modified Jablonski retains a non-processed (not thermoformed) planar shape – whereby each construction is suitable for the conveyance of gas to the patient at a therapeutic pressure.

By this construction methodology, one of ordinary skill in the art would have expected the modified Jablonski to perform equally well with Applicant's invention as the construction would yield the predictable results of an outer textile layer having an arcuate 3-D shape whilst the inner textile layer would have a flat planar shape.

Applicant has not asserted the specific construction methodology of thermoforming some components whilst not thermoforming other components of the at least one gas delivery tube provides a particular advantage, solves a stated problem, or serves a particular purpose different that providing a shaped gas delivery tube which includes a flat and planar shape of the inner textile layer as exposed to the patient's face and an arcuate 3-D shape of the outer textile layer as exposed to the ambient environment - to permit the conveyance of gas to the patient at a therapeutic pressure; thus, the specific construction methodology of thermoforming lacks criticality in its design.

As an aside, Applicant is further reminded this instant claim is effectively a product by process claim. Pursuant to MPEP 2113, "[E]ven though product-by-process claims are limited by and defined by the process, determination of patentability is based on the product itself. The

patentability of a product does not depend on its method of production. If the product in the product-by-process claim is the same as or obvious from a product of the prior art, the claim is unpatentable even though the prior product was made by a different process." In re Thorpe, 777 F.2d 695, 698, 227 USPQ 964, 966 (Fed. Cir. 1985). In this case, as the claims are apparatus claims, the act of utilizing the construction methodology of thermoforming or some other process effectively yields the same or obvious product – at least one gas delivery tube having an arcuate 3-D shape of the outer textile layer as exposed to the ambient environment and a flat and planar shape of the inner textile layer as exposed to the patient's face – to permit the conveyance of gas to the patient at a therapeutic pressure.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the therapeutic pressure PAP operational range of Jablonski to include operations of "at least 6 cm of water", as taught by Ovizinsky to be known PAP operational ranges suitable for PAP delivery, and to modify the manner of manufacturing of the at least one gas delivery tube of Jablonski to include thermoforming of the outer textile layer, and not thermoforming of the inner textile layer, as taught by Ovizinsky to be a known construction methodology suitable for imparting a shaped gas delivery tube which includes a flat and planar shape of the inner textile layer as exposed to the patient's face and an arcuate 3-D shape of the outer textile layer as exposed to the ambient environment - to permit the conveyance of gas to the patient at a therapeutic pressure.

As to Claim 2, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) – including components 46 and 34 as integrally connected ("In addition, in this exemplary embodiment, support members 34 are formed as an integral part of urethane layer

46."Para 0031; and "bottom surface member 30 comprises a two layer structure in the form of a urethane backed material and includes a fabric layer 44 attached to a urethane layer 46." Para 0030) – has different material properties (based on dimensional shape) than the outer textile layer (40) – including component 42 as attached ("top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030).

Should Applicant respectfully disagree with Primary Examiner's assessment of "differential material properties" based on dimensional shape, extrinsic evidence of Fiori (2008/0047559), published approximately ten (10) years prior to the instant effective filing date, clearly states the concept of modifying material properties "may be achieved by the use of different materials ***or*** *the use of materials having a different thickness." (Para 0027). Consequently, although the modified Jablonski, specifically Jablonski discloses what appears to be the same fabric layer and same urethan layer for both the inner textile layer and the outer textile layer, the thickness of the inner textile layer to include the integral component 34 yields a different thickness and thus different material properties as compared to the material properties of the outer textile layer.*

As to Claim 5, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) has a substantially planar cross section (best seen Figure 4).

As to Claim 6, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) is floppy – in the absence of the additional structural components of 46/34 as integrally connected ("In addition, in this exemplary embodiment, support members 34 are formed as an integral part of urethane layer 46."Para 0031; and "bottom surface member 30 comprises a two layer structure in the form of a urethane backed material and includes a fabric

layer 44 attached to a urethane layer 46." Para 0030); whilst, in the presence of the additional structural components 46/34 as integrally connected the planar shape and spacing between the outer textile layer (40) is reinforced.

As to Claim 7, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44 via 30) is secured ("Fluid delivery housing 20 includes a top surface member 28 and a bottom surface member 30, which are joined to one another along the outer perimeters thereof to form an airtight internal chamber 32." Para 0024) to the outer textile layer (40 via 28), wherein the outer textile layer (40) imparts tension (via frictional engagement and desire to retain bonded condition – "Fluid delivery housing 20' is then constructed by placing top surface member 28 on bottom surface member 30 and adhering the outer perimeters thereof together, such as by using a die press that melts the urethane material of the two members together or by using a suitable adhesive. Also, during such construction, the top of each support member 34 may be adhered to top surface member 28 by a suitable adhesive or some other suitable process." (Para 0031) onto the inner textile layer (44).

As to Claim 9, the modified Jablonski, specifically Jablonski discloses the at least one gas delivery tube (20x) is configured to extend continuously (best seen Figure 2) from a right side of patient, along the parietal bone, to the left side of the patient.

As to Claim 10, the modified Jablonski, specifically Jablonski discloses the at least one gas delivery tube (20x) includes an air supply conduit (10, via 6, "Patient circuit 6 is structured to communicate the flow of breathing gas from pressure generating device 4 to patient interface device 8, and typically includes a gas delivery conduit or tube coupled to fluid coupling conduit 10." Para 0017), wherein when the patient interface (Figures 1 and 2) is worn on by the

patient the air supply conduit (10) is located adjacent the parietal bone (best seen Figures 1 and 2).

As to Claim 12, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) is joined to the outer textile layer (40) at a joint (left or right of Figure 4); yet, **does not expressly disclose** the "width of the joint is varied along a length of the at least one gas delivery tube.

In operation, the modified Jablonski, specifically Jablonski discloses the pathway (32) is bounded by a particular volume based on the shape of the at least one gas delivery tube (20x) as retained through a frictional engagement to retain a bonded condition ("Fluid delivery housing 20' is then constructed by placing top surface member 28 on bottom surface member 30 and adhering the outer perimeters thereof together, such as by using a die press that melts the urethane material of the two members together or by using a suitable adhesive. Also, during such construction, the top of each support member 34 may be adhered to top surface member 28 by a suitable adhesive or some other suitable process." (Para 0031).

In light of this fact, the decision to modify the width of the joint along the length of the at least one gas delivery tube would be obvious to try choosing from a finite number of identified and predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability to convey the desired amount of volume, pressure, and flow along the length of the at least one gas delivery tube.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the joint width along the length of the at least one gas delivery tube of the modified Jablonski, a

known result effective variable in order to convey the desired amount of volume, pressure, and flow along the length of the at least one gas delivery tube.

As to Claim 13, the modified Jablonski, specifically Jablonski discloses the outer textile layer (40) has an interior surface (42, "top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030), wherein the interior surface (42) of the outer textile layer (40) has a positive curvature between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4).

As to Claim 14, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) has zero curvature between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4).

As to Claim 15, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) and the outer textile layer (40) are joined without stitching ("Fluid delivery housing 20 includes a top surface member 28 and a bottom surface member 30, which are joined to one another along the outer perimeters thereof to form an airtight internal chamber 32." Para 0024; "Fluid delivery housing 20' is then constructed by placing top surface member 28 on bottom surface member 30 and adhering the outer perimeters thereof together, such as by using a die press that melts the urethane material of the two members together or by using a suitable adhesive. Also, during such construction, the top of each support member 34 may be adhered to top surface member 28 by a suitable adhesive or some other suitable process." Para 0031).

As to Claim 17, the modified Jablonski, specifically Jablonski discloses the inner textile layer (44) and the outer textile layer (40) are joined at a first edge by a first joint (one of left or

right of Figure 4) and at a second edge by a second joint (other of left or right of Figure 4), and spaced (via 42/46 **or** 32 via 34) from each other between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4), wherein rigidizers are not utilized along the inner textile layer between the first joint (one of left or right of Figure 4) and the second joint (other of left or right of Figure 4). In the instance of the spacing **at the explicit joints** (left or right of Figure 4) there is no rigidizer as the spacing is achieved by (42/46). Thus, this construction analysis meets the limitations of the claims. Primary Examiner notes Applicant has not recited "along *the entire length* of the inner textile layer between the first joint and the second joint". Even if Applicant were to recite "along the length" Primary Examiner notes lateral regions 32 proximate the joints which does not contain a rigidizer. Thus, to effectively preclude any rigidizier, it appears the recitation should be "along the *entire* length" to overcome this analysis.

As to Claim 18, the modified Jablonski, specifically Jablonski discloses during use and pressurized by air the inner textile layer (44) expands away from the outer textile layer (40). In operation, the pathway (32) is bounded by a particular volume based on the shape of the at least one gas delivery tube (20x) as retained through a frictional engagement to retain a bonded condition ("Fluid delivery housing 20' is then constructed by placing top surface member 28 on bottom surface member 30 and adhering the outer perimeters thereof together, such as by using a die press that melts the urethane material of the two members together or by using a suitable adhesive. Also, during such construction, the top of each support member 34 may be adhered to top surface member 28 by a suitable adhesive or some other suitable process." (Para 0031). In the instance whereby the pressure within the pathway (32) is greater

than the volume of the at least one gas delivery tube (20x) the limits of the frictional engagement to retain a bonded condition of the at least one gas delivery tube (20x) will be tested and may yield an expansion of the inner textile layer (44) away from the outer textile layer (40).

As to Claim 19, the modified Jablonski, specifically Jablonski discloses the outer textile layer (40) includes a conformable material (42, "top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030), wherein the conformable material (42) is thermoformed to a predetermined non-planar cross sectional shape.

As to Claim 20, the modified Jablonski, specifically Jablonski discloses the at least one gas delivery tube (20x) is thermoformed in a predetermined shape (best seen Figures 1, 2, and 4) such that a first arm (one of 24A/B, "fluid delivery housing 20 includes a central portion 21 and first and second side portions 24A, 24B extending from central portion 21. As seen in FIGS. 1 and 2, first and second side portions 24A, 24B each generally have an L or dogleg shape. First rear strap member 22A is attached to (e.g., by be sewn thereto) and extends from first side portion 24A, and second rear strap member 22B is attached to (e.g., by be sewn thereto) and extends from second side portion 24B." Para 0021; **also see**: "fluid delivery housing 20 is configured to extend over the top of the patient's head, with the side portions 24A, 24B extending downwardly in front of the patient ears and forwardly over the patient cheeks." Para 0022) that extends along a first plane, and an upper portion (21, "a central portion 21" Para 0021) extends along a second plane that is substantially orthogonal to the first plane without external force or pressure.

7. Claims 3, 4, 8, 11, and 16 are rejected under 35 U.S.C. 103 as being unpatentable over Jablonski (2015/0128949) in view of Ovizinsky et al. (2014/0102456), as applied to Claim 1, and further in view of Resendes (9,689,512).

As to Claim 3, the modified Jablonski, specifically Jablonski discloses an inner textile layer (44) suitable for use to facilitate the delivery of pressurized air to the patient; yet, **does not expressly disclose** "an interior textile membrane ... the interior textile membrane is configured to permit air transfer through the interior textile membrane."

Resendes teaches an alternative construction of at least one gas delivery tube (best seen Figure 3) constructed with an outer textile layer (one of left or right of 42 in Figure 3, "the outer layer 42 may be made of a flexible fabric material such as Cordura® brand nylon fabric." Column 5, Lines 40-65) and an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), and an interior textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), wherein during use when pressurized with air, the interior textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) together with the inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) permit air transfer through the interior textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) – as

bounded by 46 ("the continuous liner 46 facilitates flow through and reparability of the flat section 22 while minimizing the leakage of preconditioned air." Column 5, Lines 40-65).

The resultant effect of the construction of Resendes is a multilayered tube suitable for the conveyance air from the source (12) to the sink (14) in a contained manner as bounded by the liner 46, whereby the exterior components of the at least one gas delivery tube (best seen Figure 3) are constructed with an outer textile layer (one of left or right of 42 in Figure 3), an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) and an interior textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the interior textile layer of the modified Jablonski to include an additional interior textile membrane as taught by Resendes to facilitate a multilayer tube with insulative properties that allows for the passage of ambient air/moisture to/from the ambient environment, whilst retaining a climate controlled interior passageway direct form the source to the sink.

As to Claim 4, the modified Jablonski, specifically Jablonski discloses an inner textile layer (44) suitable for use to facilitate the delivery of pressurized air to the patient; yet, **does not expressly disclose** "the inner textile layer comprises a conformable inner textile membrane, and the outer textile layer comprises a conformable outer textile membrane, wherein the outer textile layer is thicker than the inner textile layer."

Resendes teaches an alternative construction of at least one gas delivery tube (best seen Figure 3) constructed with an outer textile layer (one of left or right of 42 in Figure 3, "the

outer layer 42 may be made of a flexible fabric material such as Cordura® brand nylon fabric." Column 5, Lines 40-65) and an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), and a conformable inner textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65).

In light of the teachings of Resendes having an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) and conformable inner textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) to impart insulative properties to the at least one gas delivery tube, the further modification of the outer textile layer (one of left or right of 42 in Figure 3) to include a "conformable outer textile material", and further to construct the outer textile layer to be thicker/greater than the inner textile layer would be obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability of the multilayered construction of the at least one gas delivery tube to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment with sufficient flexibility as imparted by the outer textile layer and the conformable outer textile membrane for the environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the interior textile layer of the modified Jablonski to include an additional conformable inner textile membrane as taught by Resendes to facilitate a multilayer tube with insulative properties that allows for the passage of ambient air/moisture to/from the ambient environment, whilst retaining a climate controlled interior passageway direct form the source to the sink, and further to modify the outer textile layer of the modified Jablonski to include an additional conformable outer textile membrane, and finally to modify the thickness of the outer textile layer to be thicker/greater than the inner textile layer, as taught by Resendes to facilitate a multilayer tube with sufficient flexible properties to resist the rigors of the ambient environment which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

As to Claim 8, the modified Jablonski, specifically Jablonski discloses an inner textile layer (44) suitable for use to facilitate the delivery of pressurized air to the patient; yet, **does not expressly disclose** "the inner textile layer comprises an exterior textile sheet and an interior textile membrane, wherein the interior textile membrane is configured to resist the transfer of air through the exterior textile sheet."

Resendes teaches an alternative construction of at least one gas delivery tube (best seen Figure 3) constructed with an outer textile layer (one of left or right of 42 in Figure 3, "the outer layer 42 may be made of a flexible fabric material such as Cordura® brand nylon fabric." Column 5, Lines 40-65), with an inner textile layer (the combination of 44 and 46), wherein the inner textile layer the combination of 44 and 46) includes an exterior textile sheet (44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of

any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), and an interior textile membrane (46, "the continuous liner 46 facilitates flow through and reparability of the flat section 22 while minimizing the leakage of preconditioned air." Column 5, Lines 40-65), wherein the interior textile membrane (46) is configured to resist the transfer of air through the exterior textile sheet (44 on the complementary one of left or right of 42 in Figure 3).

The resultant effect of the construction of Resendes is a multilayered tube suitable for the conveyance air from the source (12) to the sink (14) in a contained manner as bounded by the liner 46, whereby the exterior components of the at least one gas delivery tube (best seen Figure 3) are constructed with an outer textile layer (one of left or right of 42 in Figure 3), an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) and an interior textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the inner textile layer of the modified Jablonski to include an exterior textile sheet and an interior textile membrane as taught by Resendes to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment, whilst retaining a climate controlled interior passageway direct form the source to the sink.

As to Claim 11, the modified Jablonski, specifically Jablonski discloses an outer textile layer (40) with a conformable sheet (42, "top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030); yet, **does not expressly disclose** "an exterior

textile sheet and an interior textile membrane, wherein a conformable sheet is sandwiched between the exterior textile sheet and the interior textile membrane."

Resendes teaches an alternative construction of at least one gas delivery tube (best seen Figure 3) constructed with an outer textile layer (one of left or right of 42 in Figure 3, "the outer layer 42 may be made of a flexible fabric material such as Cordura® brand nylon fabric." Column 5, Lines 40-65), with an inner textile layer (the combination of 44 and 46), wherein the inner textile layer the combination of 44 and 46) includes an exterior textile sheet (44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), and an interior textile membrane (46, "the continuous liner 46 facilitates flow through and reparability of the flat section 22 while minimizing the leakage of preconditioned air." Column 5, Lines 40-65), wherein the interior textile membrane (46) is configured to resist the transfer of air through the exterior textile sheet (44 on the complementary one of left or right of 42 in Figure 3).

In light of the teachings of Resendes having an inner textile layer (the combination of 44 and 46), wherein the inner textile layer the combination of 44 and 46) includes an exterior textile sheet (44 on the complementary one of left or right of 42 in Figure 3) and an interior textile membrane (46) to impart insulative properties to the at least one gas delivery tube, the further modification of the outer textile layer (one of left or right of 42 in Figure 3) to include a "an exterior textile sheet" and "an interior textile membrane", whereby the conformable membrane of the modified Jablonski is located between the "exterior textile sheet" and the "interior textile membrane" would be obvious to try choosing from a finite number of

identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability of the multilayered construction of the at least one gas delivery tube to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment with sufficient flexibility as imparted by the outer textile layer and the conformable outer textile membrane for the environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

It is noted by the Primary Examiner Resendes already teaches the use of an additional layer – albeit not "textile" – "the outer layer 42 may have a waterproof coating, such as polyurethane, configured so that moisture may pass through outer layer 42 in only one direction toward the outside of the hose assembly 10." (Column 5, Lines 40-65) in the construction of the outer textile layer (42) for the purpose of imparting moisture control in addition to durability. Hence, it would appear the additional features of layers including a conformable layer therebetween would yield additional properties of durability and flexibility to oviscape environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the outer textile layer of the at least one gas delivery tube of modified Jablonski to include additional layers as claimed to impart additional properties of durability and flexibility to oviscape environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

As to Claim 16, the modified Jablonski, specifically Jablonski discloses an inner textile layer (44) and an outer textile layer (40) suitable for use to facilitate the delivery of pressurized

air to the patient – by the joining of a conformable sheet (46 and 42, respectively, "bottom surface member 30 comprises a two layer structure in the form of a urethane backed material and includes a fabric layer 44 attached to a urethane layer 46." Para 0030 and "top surface member 28 includes a fabric layer 40 attached to a urethane layer 42." Para 0030) to form an air tight internal chamber (32, "an airtight internal chamber (32)" Abstract) wherein the inner textile layer (44) joins with the outer textile layer (40).

Yet, **does not expressly disclose** "the inner textile layer comprises an inner interior textile membrane, wherein the outer textile layer includes an outer interior textile membrane, wherein the outer interior textile membrane of the outer textile layer and the inner interior textile membrane of the inner textile layer are joined together along edges of the at least one gas delivery tube, wherein the inner interior textile membrane and the outer interior textile membrane join the inner textile layer to the outer textile layer and form an air impermeable membrane between the inner textile layer and the outer textile layer."

Resendes teaches an alternative construction of at least one gas delivery tube (best seen Figure 3) constructed with an outer textile layer (one of left or right of 42 in Figure 3, "the outer layer 42 may be made of a flexible fabric material such as Cordura® brand nylon fabric." Column 5, Lines 40-65) and an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65), and an inner textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3, "the insulative layer 44 may be made of

any suitable lightweight insulation material, such as fiberglass, Astro-foil, foil bubble wrap, and so forth." Column 5, Lines 40-65).

In light of the teachings of Resendes having an inner textile layer (one of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) and inner textile membrane (other of top/bottom 44 on the complementary one of left or right of 42 in Figure 3) to impart insulative properties to the at least one gas delivery tube, the further modification of the outer textile layer (one of left or right of 42 in Figure 3) to include a "outer textile material", and further to construct the outer textile layer to be thicker/greater than the inner textile layer would be obvious to try choosing from a finite number of identified, predictable solutions with a reasonable expectation of success, whereby success would be defined by the ability of the multilayered construction of the at least one gas delivery tube to effectively insulate the at least one gas delivery tube (Figure 3) from the ambient environment with sufficient flexibility as imparted by the outer textile layer and the conformable outer textile membrane for the environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

Therefore, it would have been obvious to one having ordinary skill in the art to modify both the inner textile layer and the outer textile layer of the at least one gas delivery tube of modified Jablonski to include additional layers of the inner interior textile membrane and the outer interior textile membrane as claimed to impart additional properties of durability and flexibility to oviscape environmental hazards which may effect the efficacy of the tubes operation through – e.g. puncturing, snags, rips, or tears.

Conclusion

8. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

b. Barlow et al. (11,590,310) shares at least one common assignee/inventor with the instant application and appears to disclose the features of a patient interface having a plenum chamber, seal forming structure and positioning and stabilizing structure wherein the method of thermoforming is performed in the construction of at least one gas delivery tube – see patent claim 19 in which the instantly claimed "outer textile layer" corresponding to the patent claimed "non-patient contacting portion ... is thermoformed; yet, **at this time there does not appear** to be double patenting as it fails to address the explicit configuration as claimed in the instant claims whereby "the outer textile layer is thermoformed to have a predetermined shape and the inner textile layer is not thermoformed". Barlow remains silent to the construction of the instantly claimed "inner textile layer" corresponding to the patent claimed "patient contacting portion" not being thermoformed. In this absence of a clear and unmistakable recitation of a preclusion to being thermoformed, Primary Examiner cannot assert double patenting.

c. Davidson et al. (12,268,815) share at least one common assignee/inventor with the instant application and appears to disclose the features of a patient interface having a plenum chamber, seal forming structure and positioning and stabilizing structure wherein the method of thermoforming is performed in the construction of at least one gas delivery tube – see patent claim 4 in which the instantly claimed "outer textile layer"

corresponding to the patent claimed "outer textile portion ... is thermoformed"; yet, **at this time there does not appear** to be double patenting as it fails to address the explicit configuration as claimed in the instant claims whereby "the outer textile layer is thermoformed to have a predetermined shape and the inner textile layer is not thermoformed". Davidson remains silent to the construction of the instantly claimed "inner textile layer" corresponding to the patent claimed "inner textile portion" not being thermoformed. In this absence of a clear and unmistakable recitation of a preclusion to being thermoformed, Primary Examiner cannot assert double patenting.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNETTE F DIXON whose telephone number is (571)272-3392. The examiner can normally be reached M-F 9-5 EST with flexible hours.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kendra D Carter can be reached at 571-272-9034. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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